Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2007-I-Q05] | Part A (Q1-Q5)

Answer: 539

1. Increasing T by nine raises rounded Celsius by five and the final Fahrenheit by nine. The success pattern is therefore periodic modulo nine.

2. For $T = 32, \ldots, 40$, final values are $32, 34, 34, 36, 36, 37, 37, 39, 39$; five inputs succeed. Through 994 there are 107 full blocks, giving 535. The six remaining temperatures match the first six residues, with four successes.

3. Total 539. No half-integer rounding ties arise in these conversions.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2002-II-Q08] | Part B (Q6-Q10)

Answer: 049

1. For positive k, only positive n matter. Such an n lies in $(2002/(k+1), 2002/k]$. Its interval length is $2002/[k(k+1)]$, at least one for $k \leq 44$, so all these values occur.

2. The choices $n = 44, 43, 42, 41$ give $45, 46, 47, 48$. The next input 40 gives 50, and monotonicity leaves no input for 49. Thus the answer is 49.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2007-I-Q07] | Part B (Q6-Q10)

Answer: 477

1. The ceiling-floor difference is zero exactly for integral logarithms and one otherwise. An integer power of $\sqrt{2}$ that is also an integer is a power of two.

2. Thus subtract $1 + 2 + \cdots + 512 = 1023$ from the sum of all integers one through one thousand: $N = 500500 - 1023 = 499477$. The remainder is 477.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2007-II-Q07] | Part B (Q6-Q10)

Answer: 553

1. The interval is $k^3 \le n < (k+1)^3 = k^3 + 3k^2 + 3k + 1$. Its multiples of k have quotients $k^2, k^2+1, \ldots, k^2+3k+3$, so there are $3k+4$.

2. From $3k + 4 = 70$, $k = 22$. Maximum quotient $22^2 + 66 + 3 = 553$.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2008-I-Q07] | Part B (Q6-Q10)

Answer: 708

1. Gaps between consecutive squares through $50^2 = 2500$ are at most ninety-nine, so every block $S_0, \ldots, S_{25}$ contains a square.

2. After that, consecutive square gaps exceed one hundred, so each later block has at most one. The squares 51^2 through 316^2 occupy 266 distinct later blocks; $317^2 > 99999$.

3. Of the 974 later blocks, $974 - 266 = 708$ are empty of squares.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2009-I-Q06] | Part B (Q6-Q10)

Answer: 412

1. On $k \le x < k+1$ for positive integer k, the image is $[k^k, (k+1)^k)$, containing $(k+1)^k - k^k$ integers. These intervals for different k are disjoint.

2. For $k = 1, 2, 3, 4$ the counts are $1, 5, 37, 369$, all below one thousand; $5^5 > 1000$ excludes larger k. Values $0 < x < 1$ only yield one, already counted. Negative x gives a negative value or a positive value below one, so no additional integer.

3. Total $1 + 5 + 37 + 369 = 412$.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2010-I-Q08] | Part B (Q6-Q10)

Answer: 132

1. The integer floor pairs are $(\pm 5, 0), (0, \pm 5), (\pm 3, \pm 4), (\pm 4, \pm 3)$. Each contributes a half-open unit square; any closed enclosing disk also contains their closures.

2. The union of closed squares is centrally symmetric about $Q = (1/2, 1/2)$. An opposite pair $X, 2Q - X$ forces every enclosing radius to be at least $|X - Q|$.

3. The greatest corner distance from Q is attained at $(4, 5)$ and its symmetric counterparts: its square is $(7/2)^2 + (9/2)^2 = 130/4$. All other corners have no greater distance, so the disk centered at Q with radius $\sqrt{130}/2$ attains the bound.

4. Thus $m + n = 130 + 2 = 132$.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2012-II-Q10] | Part B (Q6-Q10)

Answer: 496

1. With $a = \lfloor x \rfloor$, positivity of n forces $a \geq 1$. The range $a \leq x < a + 1$ gives $a^2 \leq n < a^2 + a$. Conversely every integer in this interval is attained by $x = n/a$.

2. These integer intervals are disjoint, and each contributes exactly a numbers. All intervals for $a = 1, \ldots, 31$ lie below 1000, ending at 991; the next begins at 1024.

3. The count is $1 + \cdots + 31 = 496$.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2021-I-Q08] | Part B (Q6-Q10)

Answer: 057

1. Let $F(x) = |20|x| - x^2|$, an even function. On $x \geq 0$, it rises from 0 to 100, falls to 0 at 20, then rises without bound.

2. Thus the level $F(x) = k$ has 6 roots for $0 < k < 100$, 4 at $k = 100$, 2 for $k > 100$, 3 at $k = 0$, and none for $k < 0$.

3. The two distinct levels are $c - 21, c + 21$. Twelve roots require six from each, so $0 < c - 21 < c + 21 < 100$. The integer range is $22 \leq c \leq 78$, giving 57.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2023-I-Q10] | Part B (Q6-Q10)

Answer: 944

1. Without floors the sum is $S(a) = [\sum n^2 - a\sum n]/5$. It vanishes at $a = (2\cdot 2023+1)/3 = 1349$.

2. Changing a by one changes S by $\sum n/5 = 2023\cdot 1012/5 > 400000$, whereas floors decrease the sum by less than 2023. Thus only $a = 1349$ can work.

3. For this value, $n^2 - na \equiv n(n+1) \pmod 5$, whose residues for $n \equiv 0,1,2,3,4$ are $0,2,1,2,0$. Their total through $2023 = 5\cdot 404 + 3$ is $404\cdot 5 + (2+1+2) = 2025$.

4. Because $S(1349) = 0$, the floor sum is $U = -2025/5 = -405$. This satisfies the bound and gives $a + U = 944$.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2004-I-Q12] | Part C (Q11-Q15)

Answer: 014

1. For base b, an even floor $2j$ corresponds to $b^{-2j-1} < x \leq b^{-2j}$. These disjoint intervals have total length $\sum_{j\geq 0}(1-1/b)b^{-2j} = b/(b+1)$.

2. The x- and y-conditions are independent Cartesian factors of lengths $2/3, 5/6$. Area $= (2/3)(5/6) = 5/9$, giving 14.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2007-I-Q11] | Part C (Q11-Q15)

Answer: 955

1. $b(p) = k$ exactly when $k^2 - k + 1 \le p \le k^2 + k$, a block of $2k$ integers. The blocks $k = 1, \ldots, 44$ end at 1980.

2. They contribute $2\sum_{k=1}^{44} k^2 = 58740$. Remaining 27 inputs each give 45, adding 1215. Total 59955, remainder 955.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2008-I-Q12] | Part C (Q11-Q15)

Answer: 375

1. For $k = \lceil v/15 \rceil$, successive fronts are $4(k+1)$ meters apart, giving flow $F = 250v/(k+1) \leq 3750k/(k+1) < 3750$ cars per hour.

2. A finite hour can include an initial car at its starting instant and then one per completed interval. Therefore at most 3750 cars can pass: 3751 would require 3750 gaps within an hour, contradicting $F < 3750$.

3. Choose k large enough that $F > 3749$, and start with a car at the point; then 3750 cars do pass. Thus $M = 3750$, whose quotient by ten is 375.

4. If instead whole cars per hour means the floor of the steady flow rate, its largest possible value is 3749 because $3749 < F < 3750$ is attainable. Under finite-window counting with a free initial phase it is 3750 as above. Both conventions give the requested integer quotient 375.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2009-II-Q11] | Part C (Q11-Q15)

Answer: 125

1. The interval is $m/n < k < mn$. Write $m = nq + r$, $0 \le r < n$, $q \ge 1$. The count is $mn - q - 1$, so $q(n^2 - 1) + nr = 51$; also $n \ge 2$.

2. For $n \ge 8$ the first term already exceeds fifty-one. Testing $2 \le n \le 7$ with $0 \le r < n$ leaves $(n, q, r) = (2, 17, 0), (3, 6, 1)$, giving $(m, n) = (34, 2), (19, 3)$.

3. The distinct products are $68, 57$ and their sum is 125.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2010-I-Q14] | Part C (Q11-Q15)

Answer: 109

1. Every term is nondecreasing in n. It therefore suffices to exhibit adjacent integers where the bound changes.

2. At $n = 109$, the products have floor logarithms 2 for $k = 1, \ldots, 9$, 3 for $10, \ldots, 91$, and 4 for $92, \ldots, 100$. Their sum is $18 + 246 + 36 = 300$.

3. At $n = 110$, the threshold for 10^4 shifts to $k = 91$, so the sum is $18 + 243 + 40 = 301$. Monotonicity proves the maximum is 109.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2011-II-Q15] | Part C (Q11-Q15)

Answer: 850

1. On a unit interval $[n, n+1)$, the right side must be an integer. For $n = 5, \ldots, 14$, the values $P(n)$ are $1, 9, 19, 31, 45, 61, 79, 99, 121, 145$; only $n = 5, 6, 13$ work, with roots $1, 3, 11$.

2. The polynomial increases throughout the sample interval. For each working n with $P(n) = j^2$, success persists from n until the root of $P(x) = (j+1)^2$, namely $[3 + \sqrt{45 + 4(j+1)^2}]/2$. These cutoffs all lie before $n + 1$.

3. The three interval lengths are $(\sqrt{61} - 7)/2, (\sqrt{109} - 9)/2, (\sqrt{621} - 23)/2$. Divide their sum by ten to obtain $(\sqrt{61} + \sqrt{109} + \sqrt{621} - 39)/20$.

4. The endpoint 15 has probability zero. The required sum is $61 + 109 + 621 + 39 + 20 = 850$.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2015-I-Q14] | Part C (Q11-Q15)

Answer: 483

1. On a unit strip $j \le x < j+1$, the floor is constant $k = \lfloor \sqrt{j} \rfloor$. Its trapezoid area is $k(2j+1)/2$. Thus the fractional part of cumulative area changes by $1/2$ exactly when k is odd.

2. Block k consists of $k^2 < n \le (k+1)^2$ and has $2k+1$ values. Starting at $A(1) = 0$, the status at block starts cycles modulo 4: integer for $k \equiv 0, 1$, half-integer for $k \equiv 2, 3$.

3. For four successive blocks $k = 4r-3, 4r-2, 4r-1, 4r$, the counts of integer areas are $4r-3, 0, 4r, 8r+1$. For $r = 1, \ldots, 7$, these sum to $\sum(16r-2) = 434$, covering through $n = 841$.

4. Block $k = 29$ adds 29 values through 900; block 30 adds none through 961; the partial block 31 begins with an integer at $n = 962$ and adds 20 through 1000. Total $434 + 29 + 20 = 483$.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2020-II-Q14] | Part C (Q11-Q15)

Answer: 010

1. On $[j, j+1)$, $f(j+t) = (j+t)t$ increases bijectively from 0 to the range $[0, j+1)$. In particular integer endpoints cannot solve the target equation.

2. The inverse of 17 has one point in each interval $(n, n+1)$ for $n \geq 17$. Taking another inverse produces one point in every $(k, k+1)$ with $k \geq n$. A third inverse in $(j, j+1)$ exists exactly for $17 \leq n \leq k \leq j$.

3. For fixed j, the count is $\binom{j-15}{2}$. Sum for $17 \leq j \leq 2019$ to obtain $N = \binom{2005}{3} = 1341349010$. The remainder is 10.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2024-I-Q12] | Part C (Q11-Q15)

Answer: 385

1. Every solution lies in $[0,1]^2$. Put $h(u)=4||u-1/2|-1/4|$ on $[0,1]$. Its values at $0,1/4,1/2,3/4,1$ are $1,0,1,0,1$. Thus $y=F(x)=h(|\sin 2\pi x|)$ has sixteen monotone full-height branches, while $x=G(y)=h(|\cos 3\pi y|)$ has twenty-four full-width branches. The only boundary solution is $(1,1)$.

2. To justify branch-intersection uniqueness, we prove $|F\prime(x)G\prime(y)|=96\pi^2|\cos 2\pi x\sin 3\pi y|>1$ at every interior solution. Such solutions avoid all branch endpoints. If $|\cos 2\pi x|\le 1/20$, then $|\sin 2\pi x|\ge 399/400$, hence $y\ge .99$. It follows that $x\ge 1-18\pi^2(.01)^2>.982$, forcing $|\cos 2\pi x|>.99$, a contradiction.

3. If the derivative product were at most one, at least one factor $|\cos 2\pi x|,|\sin 3\pi y|$ would be at most $1/20$, since $96\pi^2/400>1$. The previous argument excludes the first, so $|\sin 3\pi y|\le 1/20$ and $x\ge .99$. Set $a=1-x$, $b=1-y$. Then $0<a\le .01$ and $b=4\sin 2\pi a\le .08\pi<.252$. The sine bound now forces $0<b<.01$ (the only nearby zero of $\sin 3\pi y$ is at $y=1$). The exact equations are $b=4\sin 2\pi a$ and $a=4(1-\cos 3\pi b)$.

4. Define $H(b)=4(1-\cos 3\pi b)-\arcsin(b/4)/(2\pi)$. On $[0,.01]$, $H\prime\prime(b)=36\pi^2\cos 3\pi b-b/[128\pi(1-b^2/16)^{3/2}]>0$. We have $H(0)=H(b)=0$ for the noncorner solution, so strict convexity gives $H\prime(b)>0$. This says $12\pi\sin 3\pi b>1/(8\pi\cos 2\pi a)$, precisely that the derivative product exceeds one, a contradiction. The claimed inequality follows.

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2024-I-Q12] | Part C (Q11-Q15) (continued / 续)

5. For oppositely oriented branches uniqueness follows directly by monotonicity. For equally oriented branches, write the first as $x = F^{-1}(y)$: the difference $G(y) - F^{-1}(y)$ has the same nonzero crossing sign at every zero because the derivative product is greater than one. It cannot have two such crossings. Boundary endpoint signs guarantee one interior crossing in each of the $16 \cdot 24$ branch-pair rectangles, except that the final pair shares $(1, 1)$. There is also an interior crossing for that pair: near $b = 0$, $H(b) < 0$ because $H\prime(0) < 0$, while the opposite rectangle endpoint has positive sign. Therefore there are 384 interior intersections plus the corner, giving 385.

Source / 来源

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2025-I-Q11] | Part C (Q11-Q15)

Answer: 259

1. All intersections have $-1 \le y \le 1$ and $0 \le x \le 34$. Rising pieces give $34y^2 - y - 4k = 0$; falling pieces give $34y^2 + y - (4k+2) = 0$. A root is valid exactly when its y value is in the corresponding piece range.

2. The rising quadratics have both roots valid for $k = 0, \ldots, 8$, each with sum $1/34$. The falling ones have both roots valid for $k = 0, \ldots, 7$, each with sum $-1/34$; for $k = 8$ only $(-1 + 5\sqrt{185})/68$ is valid.

3. The total is $(1 + 5\sqrt{185})/68$, so $a + b + c + d = 1 + 5 + 185 + 68 = 259$.

Source / 来源